

GeoMatch: photo geolocation as visual retrieval

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1 Task

Predict the GPS coordinate of a photo from twelve European countries, scored by median haversine error. The rules permit *one* model of at most 5,000,000 parameters, no pretrained weights, and no data beyond the 11,758 labelled training images. Evaluation is on 2,400 held-out photos.

2 Why retrieval

The model predicts a location by matching the query against the training set and copying the coordinate of the best match. This does *not* make the output continuous — predictions are restricted to the 11,758 training coordinates, which is a real quantisation. It is simply a cheap one: the geographically nearest training photo to a held-out photo is a median of **3.1 km** away, and within 50 km for **99.8%** of queries. That ceiling is an order of magnitude below the error we actually achieve, so discretisation is not the binding constraint, whereas a geo-cell classifier is bounded by its cell size.

The real argument for retrieval is that it removes the coordinate regression problem entirely: the encoder only has to learn *which photos were taken near each other*, a metric-learning objective that a small from-scratch network can be supervised on densely, using every pair in a batch.

3 Model (Fig. 1)

A single RegNetY-400MF trunk (`weights=None`) is shared across three views: the full frame at 512^2 and aligned left/right crops at 256^2 , which preserve detail that a single downscaled frame destroys.

Global branch. Multi-scale GeM pooling over stages 2–4, fused across views, through an identity-initialised 384×384 layer, L2-normalised to a 384-D descriptor g .

Local branch. Stage-4 features average-pooled to 4×4 per view (48 tokens), projected $440 \rightarrow 64$, L2-normalised to tokens T .

For query q and bank image b , the shortlist is the cosine top-200 of $\langle g_q, g_b \rangle$, rescored by a symmetric MaxSim late interaction

$$s(q, b) = 0.2 \langle g_q, g_b \rangle + 0.8 \cdot \frac{1}{2} \left(\overline{\max_j T_q T_b^\top} + \overline{\max_i T_q T_b^\top} \right)$$

where $\overline{\max_j}$ averages the per-token maxima over one axis. The top-1 candidate’s coordinate is the prediction.

4 Training

Two stages, evaluated on the official 5-fold split (train four folds, predict the fifth, fixed epoch count, no checkpoint selection).

1. BYOL pretraining of the trunk (negative-free, momentum target and predictor), labels unused, 160 epochs.

2. Country-aware retrieval fine-tuning, 40 epochs, AdamW (8×10^{-5} trunk, 6×10^{-4} heads), batch 64, EMA 0.999. Each sampled group is an anchor, two geographic positives (≤ 50 km, drawn by descriptor similarity from the 64 nearest), and one *same-country* hard negative 60–700 km away, mined from the current EMA encoder and refreshed every epoch. Anchors from the weakest countries are oversampled (DE 3.0 $\times$, FR 2.8, PL 2.0, IT/ES 1.7, SE 1.4, GB 1.3). The objective is 0.65 retrieval + 0.2 classification regulariser; the retrieval term mixes global listwise/triplet (0.35) with local listwise/hinge (0.65).

5 Results

pooled 5-fold OOF	n	median (km)	≤ 50 km
From-scratch retrieval baseline	1	89.2	45.3%
+ BYOL + country-aware, 22 ep	1	57.7	48.8%
+ 40 epochs (shipped recipe)	3	56.8 $\pm$ 1.1	48.8%
(<i>non-compliant</i>) 3-model ensemble	1	52.0	49.7%
(<i>non-compliant</i>) 6-model ensemble	1	49.7	50.1%

Table 1: n is the number of independently seeded 5-fold runs. The shipped recipe’s three seeds gave 55.60 / 57.12 / 57.71 km; we report their mean. **Best observed is 55.60 km**, but that seed was not selected for submission — the submitted model uses the config’s default seed, which was never evaluated. Ensembles are listed only to document that the gain is real and was rejected: the rules require a single ≤ 5 M-parameter model.

Self-supervised pretraining is the one large win (−31 km). Extending 22 to 40 epochs is within seed noise (± 1.1 km across three seeds), so we do not claim it as an improvement; 40 epochs was kept because the schedule was fixed in advance.

6 Where the error comes from

Splitting every query into *located* (≤ 50 km), *mis-ranked* (a ≤ 50 km bank image was in the top-200 but not returned first) and *never retrieved* (no ≤ 50 km image in the top-200) separates the two failure modes (Fig. 2).

	located	mis-ranked	never retrieved
Pooled	49.1%	30.3%	20.7%
Iceland	82.9%	15.2%	1.9%
France	25.2%	32.7%	42.1%
Germany	14.2%	40.6%	45.2%

Table 2: Ranking is the larger loss pooled, but for Germany and France the descriptor fails to retrieve any nearby image at all for $\sim 45\%$ of queries. Both stages fail there; a better reranker alone could not fix it.

This is why we stopped working on decoding. Germany and France carry the whole remaining gap (median 505 and 570 km against < 220 km everywhere else), and roughly half of their failures are already lost before any reranking happens.

GeoMatch: Single-Model Visual Retrieval Architecture

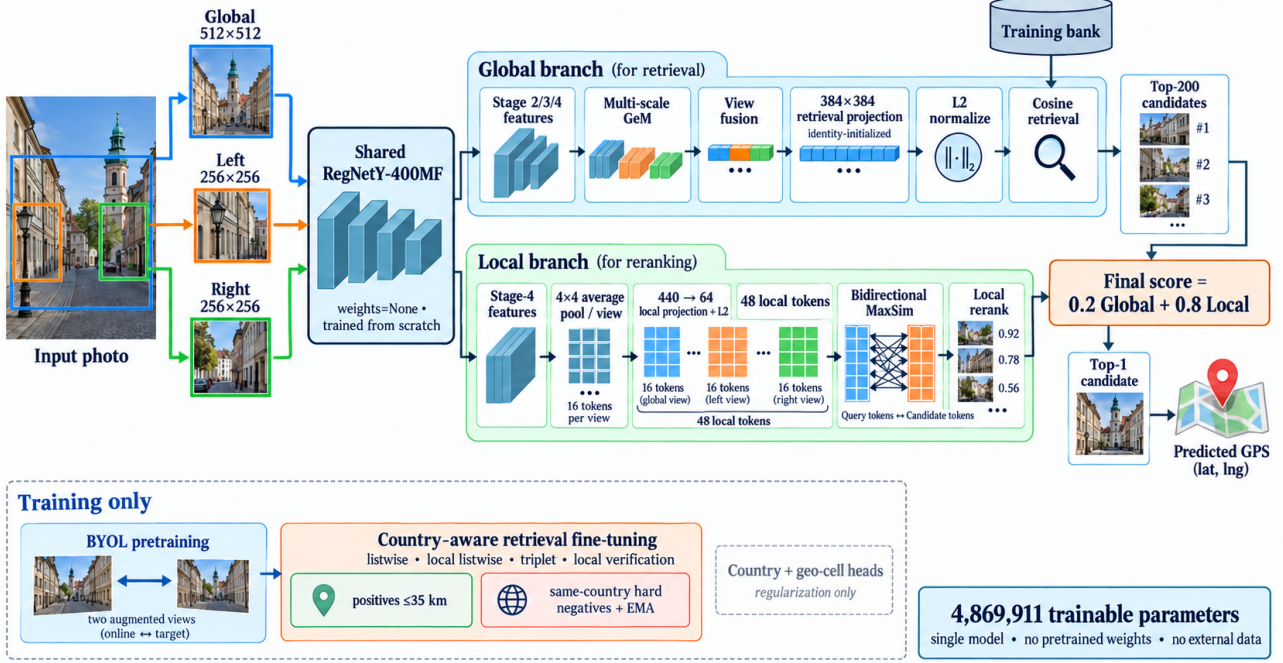


Figure 1: One from-scratch RegNetY-400MF trunk, shared across a global view and two crops, feeds a global-descriptor retrieval branch and a local-token late-interaction reranker. The classification heads are a training regulariser and are not used at inference. 4,869,911 parameters.

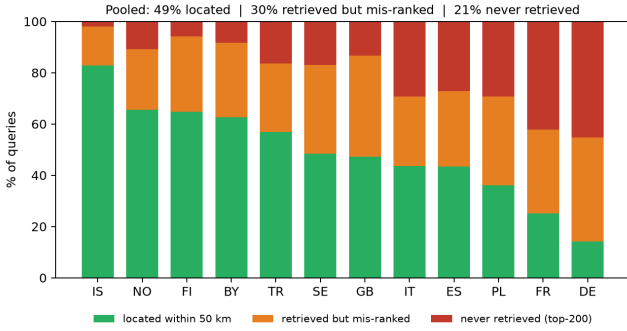


Figure 2: Error decomposition per country, sorted by success rate.

7 What did not work

Decode-time repairs on frozen descriptors (5-fold evidence). A learned candidate reranker was tried four times (linear, residual, pairwise, and one gated on a held-out inner fold) and degraded the crossfit every time; the inner-fold gate passed only because the probe fold was held out from the reranker but not from the *encoder*. Kernel-density / spatial-consensus decoding reached 72km against 57.7. Restricting the shortlist by predicted country cost 4km pooled (country accuracy is only 68%, and a wrong gate is unrecoverable). Decoding from the auxiliary geo-cell head instead of retrieval gave DE 490 / FR 523 km against retrieval’s 534 / 551 — equally lost, which is the strongest evidence that DE/FR location is simply not encoded.

Encoder variants (single-fold evidence, weaker). Adding a random-resized-crop to the global view was worse at every matched epoch on fold 0 (e.g. 63.7 vs 58.1km at epoch 32); it discards the horizon and scene layout the descriptor needs. Raising the local grid from 4×4 to 8×8 was *not* clearly worse (58.3 vs 60.7km at

epoch 40) — it is within fold-0 noise, and we kept 4×4 as the simpler option rather than because the variant failed. Extending that run to 48 epochs drifted back to 61.8km. We flag these as suggestive only: the shipped recipe’s own fold-0 median wanders between 58.1 and 63.4km over epochs 20–40, so single-fold differences under ~ 3 km carry no signal.

8 Limitations

The dominant one is Germany and France, and the evidence above says it is a representation failure, not a decoding one: a ≤ 5 M-parameter from-scratch encoder at 512px does not resolve the fine, localised cues (signage, plates, micro-architecture) that separate suburban scenes repeating over hundreds of kilometres. Country-aware oversampling and same-country hard negatives target exactly this and moved every other weak country but not these two. Secondary: the failed-variant evidence is single-fold; and the figures here are regenerated from cached out-of-fold predictions that are not checked into the repository.

9 Submission

The submitted model is the recipe above retrained once on all 11,758 images with no held-out fold, initialised from a BYOL trunk, using the configuration’s default seed. It has 4,869,911 parameters, trains in ≈ 67 min on one RTX 4000 Ada, and writes `predictions.csv` (2,400 rows) in ≈ 5 min.